GORENSTEINNESS OF A FLAT FINITE PRESENTATION MORPHISM IS CHARACTERISED BY THE EXISTENCE OF A GORENSTEIN QUASI-FINITE MORPHISM TO AFFINE SPACE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0C09 (sources/stacks-duality/pinned-original-source.tex line 6768).

Allowed standard prerequisites (editorial).

- Existence of a quasi-finite morphism to affine space over the base. Algebra, Lemma labelled algebra-lemma-quasi-finite-over-polynomial-algebra, cited at proof lines 6790-6794. Produces the pair (U, g) and sets the value of d.
- Openness of flatness and spreading out of flat finite presentation data. Algebra, Theorem labelled algebra-theorem-openness-flatness, cited at proof lines 6816-6820, and Algebra, Lemma labelled algebra-lemma-flat-finite-presentation-limit-flat, cited at proof lines 6820-6824. Allows assuming the polynomial algebra map flat and reduces to a Noetherian base.
- Base change of the Gorenstein locus for flat finite type morphisms. Same chapter, Lemma labelled lemma-base-change-gorenstein, cited at proof lines 6824-6828. Justifies the reduction to the Noetherian case.
- The upper shriek of the affine space projection and the Noetherian Gorenstein criterion. Same chapter, Lemmas labelled lemma-upper-shriek-composition and lemma-shriek-affine-line, cited at proof lines 6834-6838, and Lemma labelled lemma-affine-flat-Noetherian-gorenstein, cited at proof lines 6838-6841. Supplies f^!O_S = g^!O_{A^n_S}[n] and concludes the equivalence.

Source reference. Original lemma 00QE: sources/stacks-chow/tags.txt, source lines 964–964. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6793 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original theorem 00RC: sources/stacks-chow/tags.txt, source lines 998–998. Cross-chapter standard prerequisite located in the project tag list

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staged in the local snapshot; invoked at proof line 6813 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02JO: sources/stacks-chow/tags.txt, source lines 3343–3343. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6816 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0C07: sources/stacks-duality/pinned-original-source.tex, source lines 6614–6614. Same-chapter pointer into the author environment lemma at source line 6614; invoked at proof line 6822 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ATX: sources/stacks-duality/pinned-original-source.tex, source lines 4161–4161. Same-chapter pointer into the author environment lemma at source line 4161; invoked at proof line 6837 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA1: sources/stacks-duality/pinned-original-source.tex, source lines 4609–4609. Same-chapter pointer into the author environment lemma at source line 4609; invoked at proof line 6838 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0C08: sources/stacks-duality/pinned-original-source.tex, source lines 6722–6722. Same-chapter pointer into the author environment lemma at source line 6722; invoked at proof line 6840 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is flat and locally of finite presentation. Let $x \in X$ with image $s \in S$. Set $d = \dim_x(X_s)$. The following are equivalent*

(1) *f is Gorenstein at x,*
(2) *there exists an open neighbourhood $U \subset X$ of x and a locally quasi-finite morphism $U \to \mathbf{A}^d_S$ over S which is Gorenstein at x,*
(3) *there exists an open neighbourhood $U \subset X$ of x and a locally quasi-finite Gorenstein morphism $U \to \mathbf{A}^d_S$ over S,*
(4) *for any S-morphism $g : U \to \mathbf{A}^d_S$ of an open neighbourhood $U \subset X$ of x we have: g is quasi-finite at $x \Rightarrow g$ is Gorenstein at x.*

In particular, the set of points where f is Gorenstein is open in X.

Proof. Choose affine open $U = \mathrm{Spec}(A) \subset X$ with $x \in U$ and $V = \mathrm{Spec}(R) \subset S$ with $f(U) \subset V$. Then $R \to A$ is a flat ring map of finite presentation. Let $\mathfrak{p} \subset A$ be the prime ideal corresponding to x. After replacing A by a principal localization we may assume there exists a quasi-finite map $R[x_1, \ldots, x_d] \to A$, see Algebra, Lemma 00QE. Thus there exists at least one pair (U, g) consisting of an open neighbourhood $U \subset X$ of x and a locally[1] quasi-finite morphism $g : U \to \mathbf{A}^d_S$.

[1] If S is quasi-separated, then g will be quasi-finite.

Having said this, the lemma translates into the following algebra problem (translation omitted). Given $R \to A$ flat and of finite presentation, a prime $\mathfrak{p} \subset A$ and $\varphi : R[x_1, \ldots, x_d] \to A$ quasi-finite at $\mathfrak{p}$ the following are equivalent

(a) $\mathrm{Spec}(\varphi)$ is Gorenstein at $\mathfrak{p}$, and
(b) $\mathrm{Spec}(A) \to \mathrm{Spec}(R)$ is Gorenstein at $\mathfrak{p}$.
(c) $\mathrm{Spec}(A) \to \mathrm{Spec}(R)$ is Gorenstein in an open neighbourhood of $\mathfrak{p}$.

In each case $R[x_1, \ldots, x_n] \to A$ is flat at $\mathfrak{p}$ hence by openness of flatness (Algebra, Theorem 00RC), we may assume $R[x_1, \ldots, x_n] \to A$ is flat (replace A by a suitable principal localization). By Algebra, Lemma 02JO there exists $R_0 \subset R$ and $R_0[x_1, \ldots, x_n] \to A_0$ such that R_0 is of finite type over $\mathbf{Z}$ and $R_0 \to A_0$ is of finite type and $R_0[x_1, \ldots, x_n] \to A_0$ is flat. Note that the set of points where a flat finite type morphism is Gorenstein commutes with base change by Lemma 0C07. In this way we reduce to the case where R is Noetherian.

Thus we may assume X and S affine and that we have a factorization of f of the form

$$X \xrightarrow{g} \mathbf{A}^n_S \xrightarrow{p} S$$

with g flat and quasi-finite and S Noetherian. Then X and $\mathbf{A}^n_S$ are separated over S and we have

$$f^!\mathcal{O}_S = g^!p^!\mathcal{O}_S = g^!\mathcal{O}_{\mathbf{A}^n_S}[n]$$

by know properties of upper shriek functors (Lemmas 0ATX and 0AA1). Hence the equivalence of (a), (b), and (c) by Lemma 0C08. □